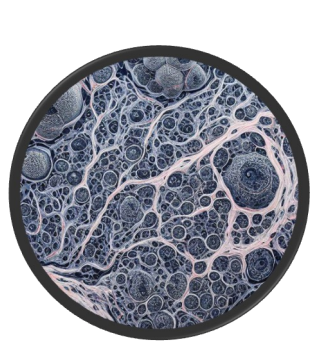


Single cell transcriptomics/ single cell RNA sequencing/ scRNAseq



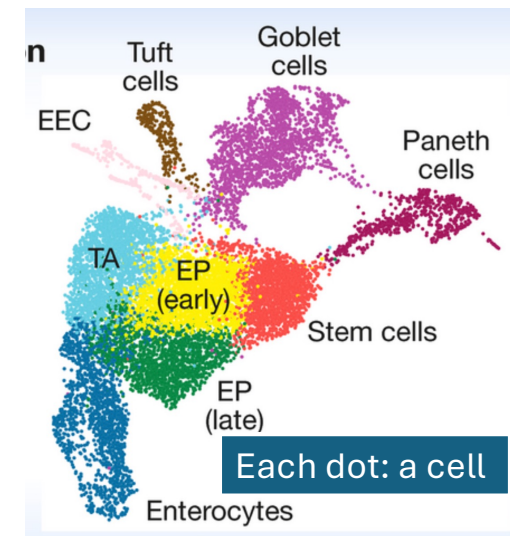
Tissue



Ind. cells

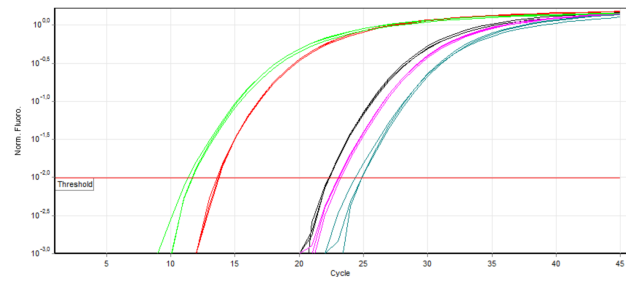


**Transcriptome
sequencing**



Luecken & Theis 2019

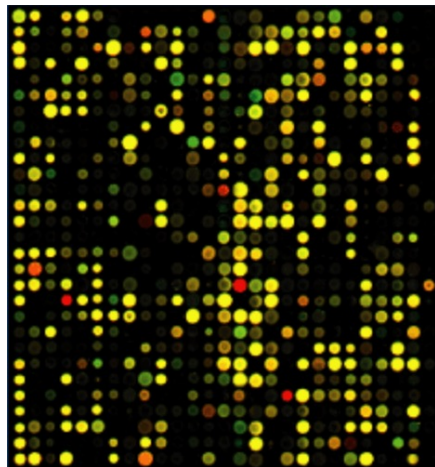
qRT-PCR



- PCR
- Sensitive
- A small number of genes/transcripts



microarray

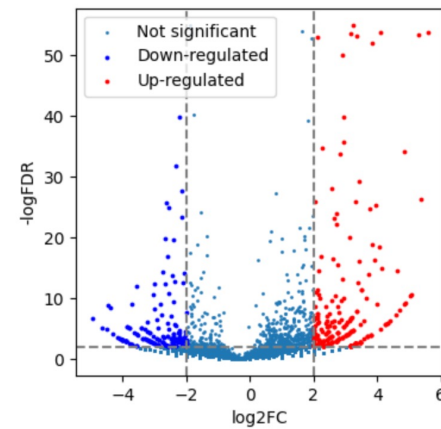


- Hybridization based
- Genome wide survey of the **transcriptome**
- Only profile predefined transcripts/genes on the chip

A **transcriptome** is the full range of messenger RNA, or mRNA, molecules expressed by an organism



Bulk RNAseq

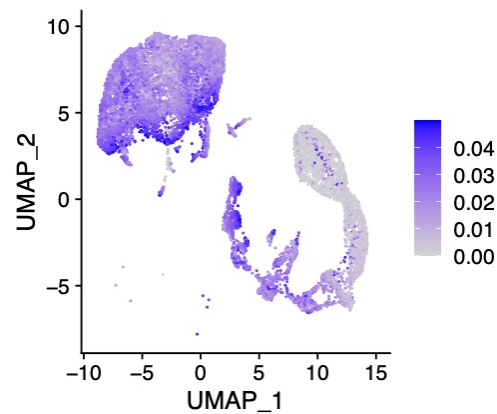


A **transcriptome** is the full range of messenger RNA, or mRNA, molecules expressed by an organism

- Full sequencing of the whole **transcriptome**



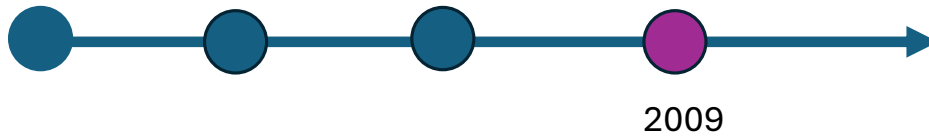
scRNAseq

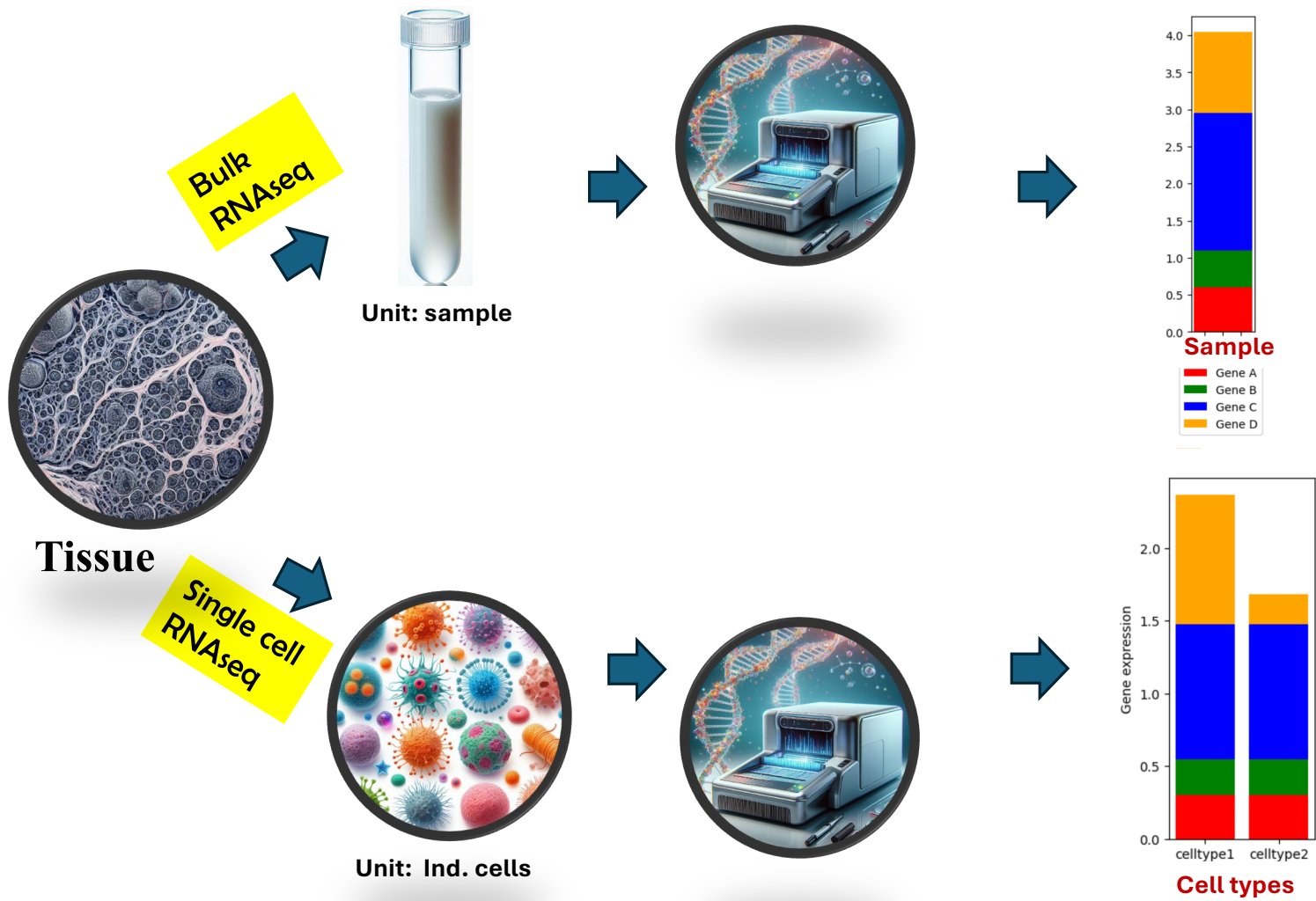


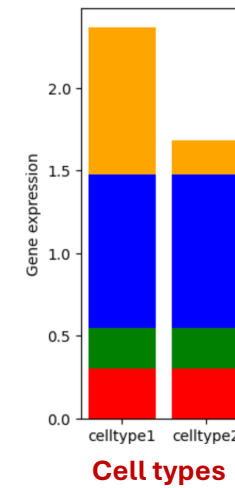
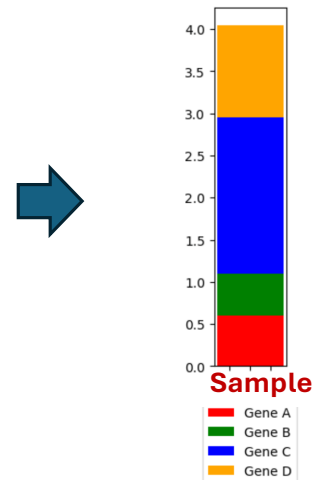
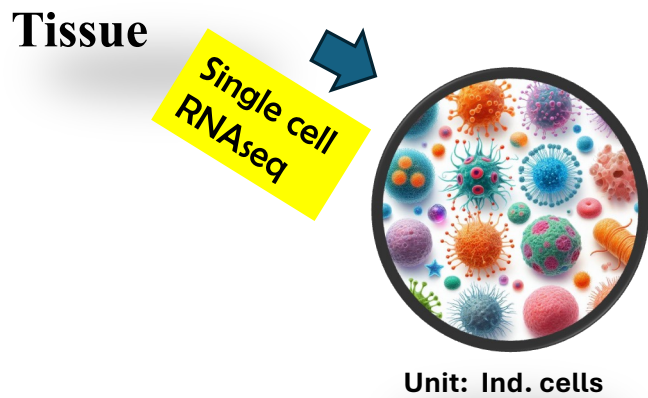
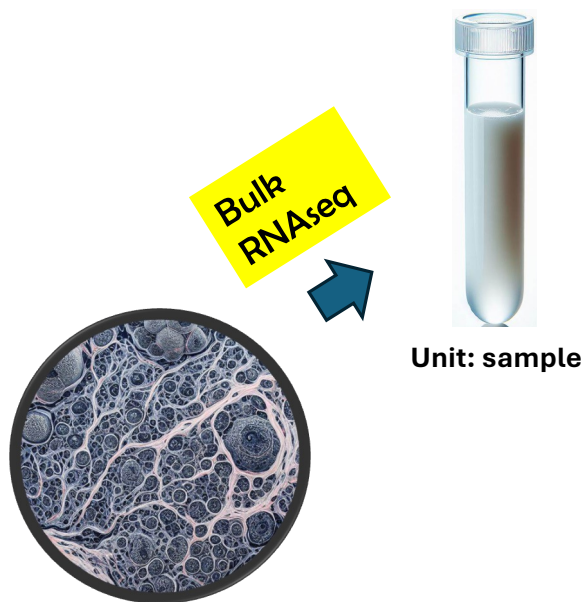
- Sequencing
- Genome wide survey of the **transcriptome**
- Individual cells

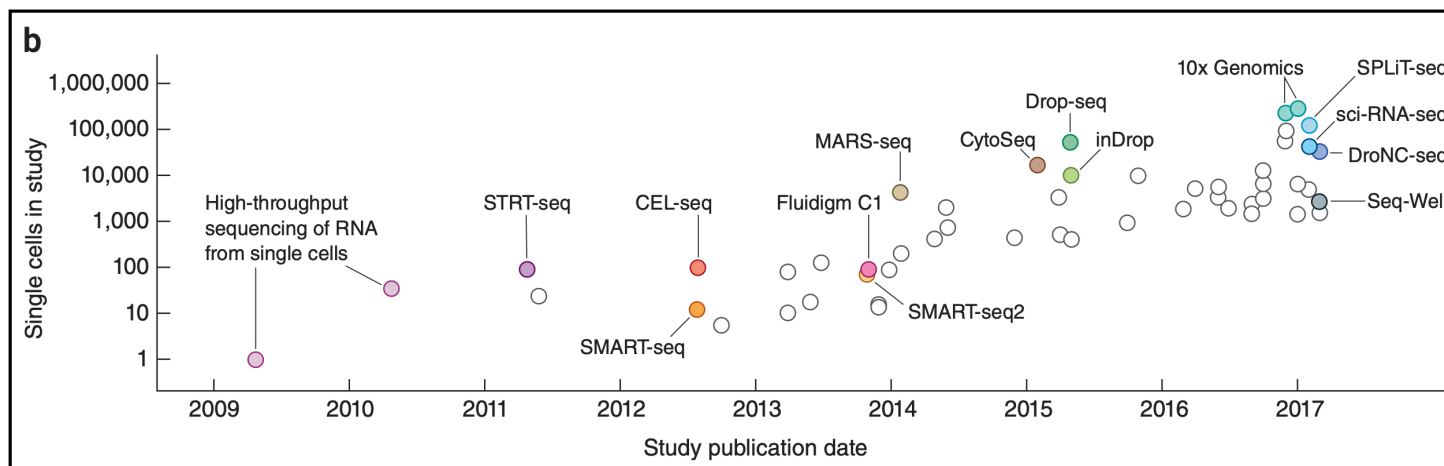


A **transcriptome** is the full range of messenger RNA, or mRNA, molecules expressed by an organism

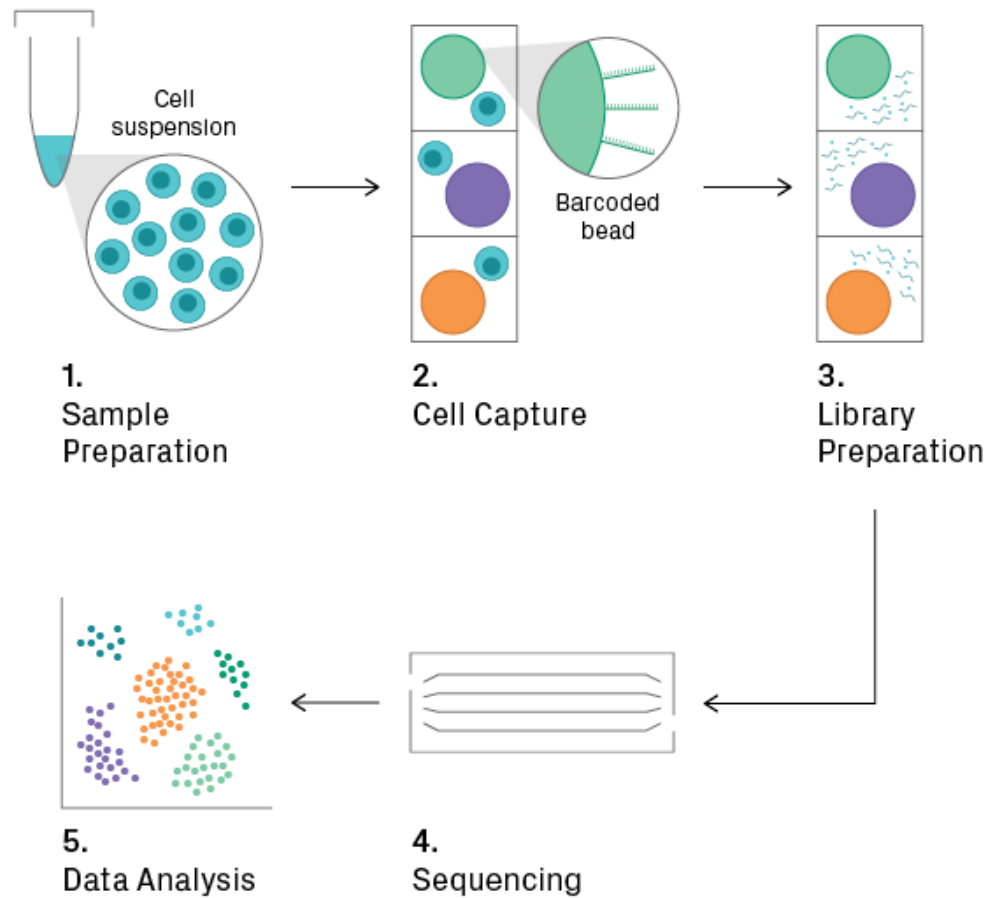




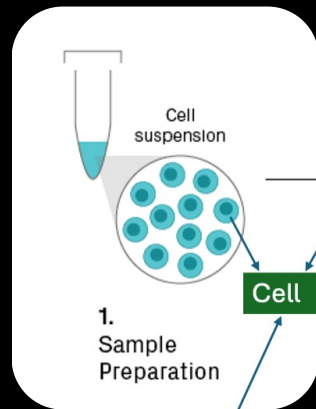


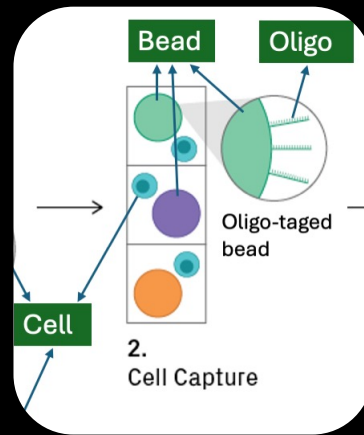


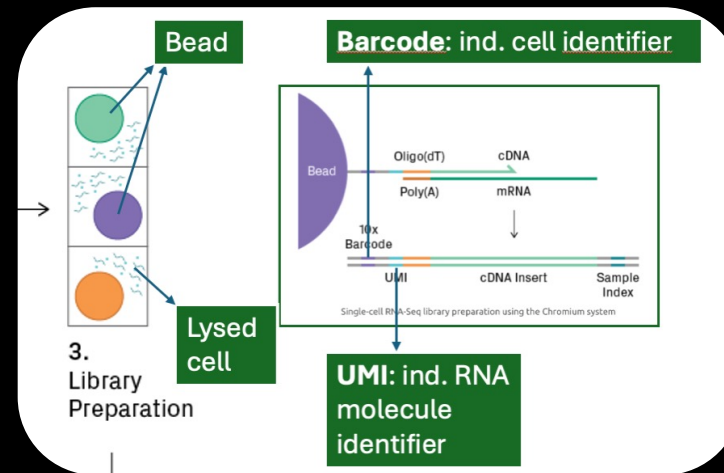
Svensson et al. 2018 Nature

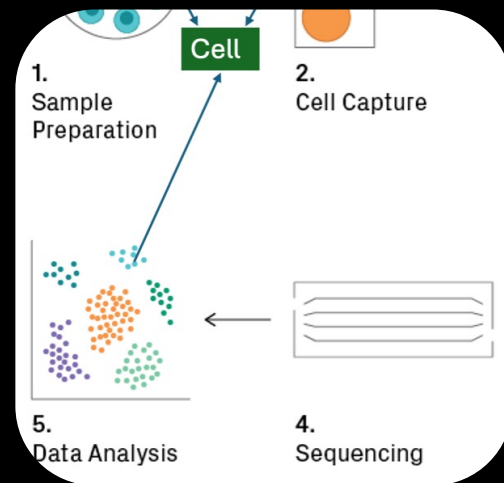


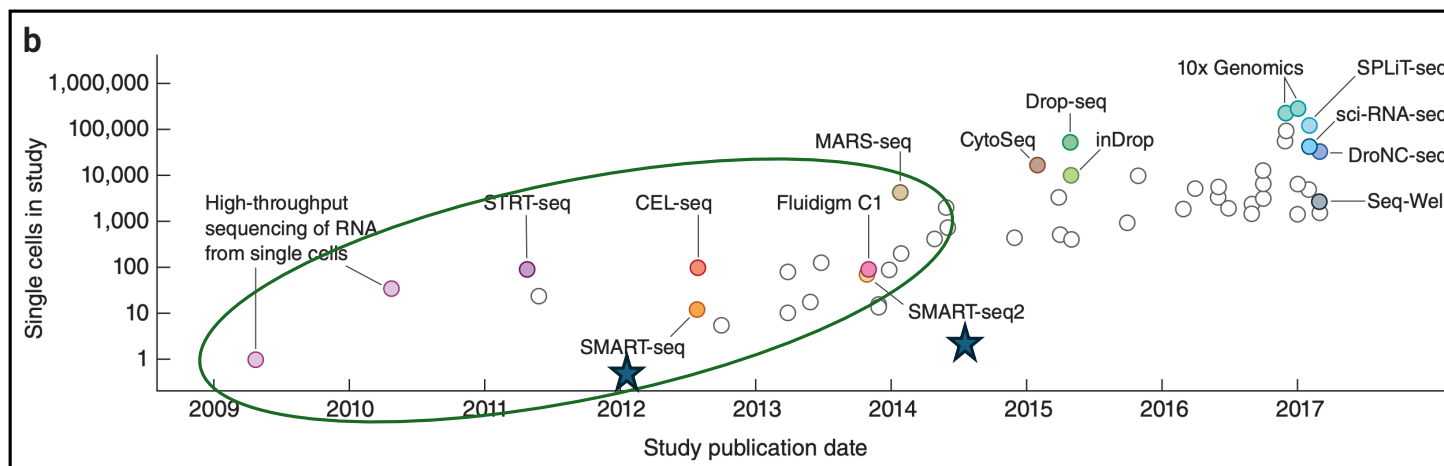
Single-cell sequencing workflow





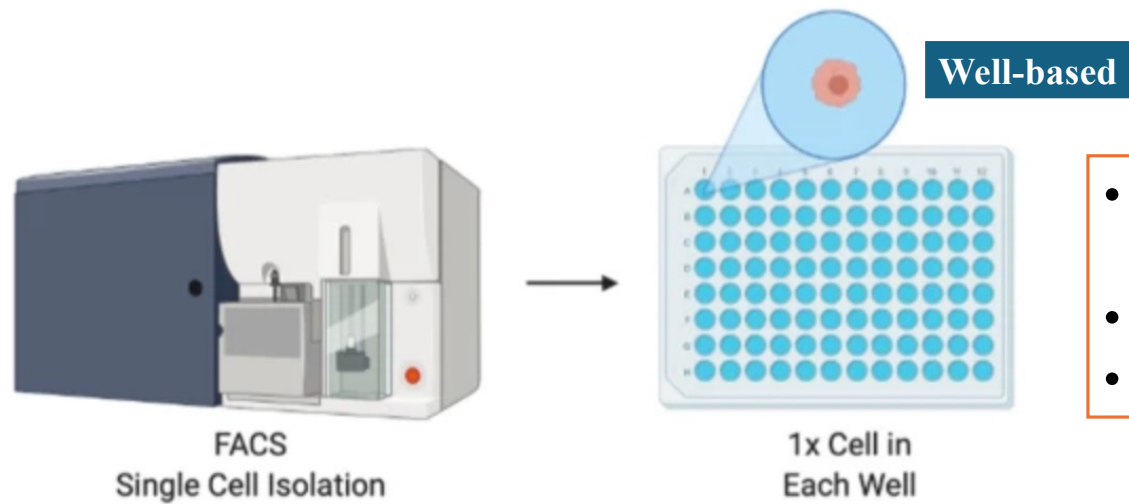






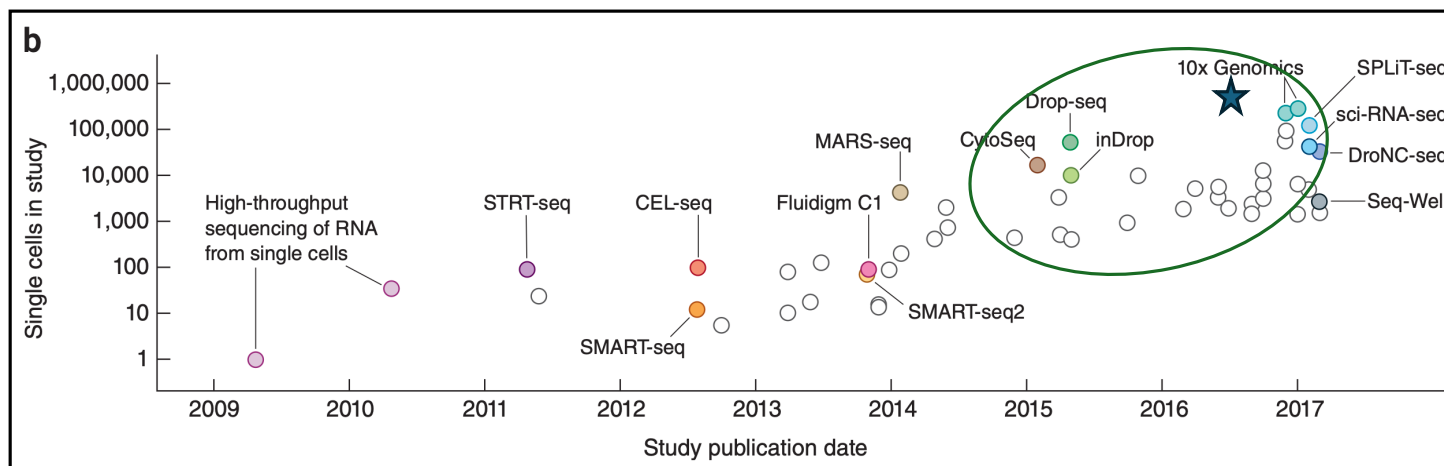
Svensson et al. 2018 Nature

SMART-seq



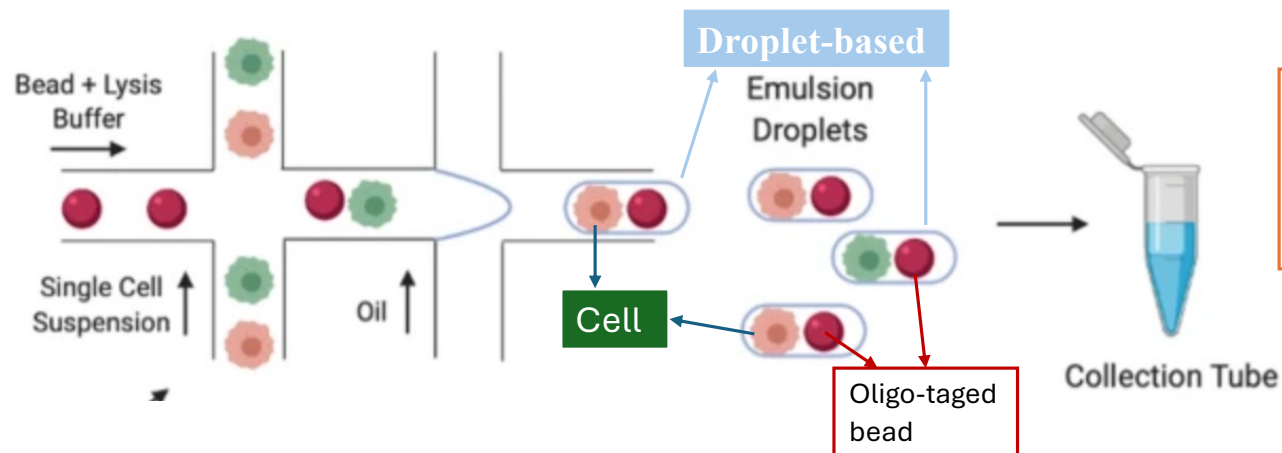
- RNA full length sequencing (good for isoform...)
- Less cells
- Lots of manual efforts

Flow cytometry



Svensson et al. 2018 Nature

10X Genomics



- 3'/5' RNA sequencing
- A large number of cells
- Lots of missing values

Microfluidic device

- ❖ What is scRNAseq
- ❖ How it works
- ❖ The evolution of scRNAseq
- ❖ The difference between bulk RNAseq and scRNAseq

Pair discussion

- Think about a situation where scRNAseq is better than bulk RNAseq

Learning goals

- ❖ The difference between bulk RNAseq and scRNAseq?

